

8.5 Problem Set

1. $\int \frac{3}{(x-1)(x+2)} dx$

$$\int \frac{A}{x-1} + \frac{B}{x+2} dx = \int \frac{1}{x-1} - \frac{1}{x+2} dx = \ln \left| \frac{x-1}{x+2} \right| + C$$

$$A(x+2) + B(x-1) = 3$$

$$\text{@ } x=1, 3=3A, A=1$$

$$\text{@ } x=-2, 3=-3B, B=-1$$

2. $\int \frac{8x-5}{3x^2-5x+2} dx$
 $(3x-2)(x-1)$

$$\int \frac{A}{3x-2} + \frac{B}{x-1} dx = \int \frac{-1}{3x-2} + \frac{3}{x-1} dx = -\frac{1}{3} \ln |3x-2| + 3 \ln |x-1| + C$$

$$A(x-1) + B(3x-2) = 8x-5$$

$$\text{@ } x=1, 3=B$$

$$\text{@ } x=\frac{2}{3}, -\frac{1}{3}A = \frac{1}{3}, A=-1$$

$$u=3x-2$$

$$du=3dx$$

$$dx = \frac{du}{3}$$

3. $\int \frac{3x^2+4x-6}{x^2-3x+2} dx$

$$= \int 3 dx + \int \frac{13x-12}{(x-2)(x-1)} dx = \int 3 dx + \int \frac{14}{x-2} + \frac{-1}{x-1} dx$$

$$A(x-1) + B(x-2) = 13x-12$$

$$\text{@ } x=1, -B=1, B=-1$$

$$\text{@ } x=2, A=14$$

$$3x + \ln \left(\frac{(x-2)^{14}}{|x-1|} \right) + C$$

4. $\int \frac{81}{x^3-9x^2} dx$

$$\int \frac{81}{x^2(x-9)} = \int \left(\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-9} \right) = \int \frac{-1}{x} - \frac{9}{x^2} + \frac{1}{x-9} dx$$

$$A(x)(x-9) + B(x-9) + Cx^2 = 81$$

$$x=0, -9B=81, B=-9$$

$$x=9, 81C=81, C=1$$

$$81 = Ax^2 + Cx^2, A=-1$$

$$A+C=0$$

$$= -\ln|x| + \frac{9}{x} + \ln|x-9| + C$$

$$= \frac{9}{x} + \ln \left| \frac{x-9}{x} \right| + C$$

5. $\int \frac{x^3 - 10x^2 + 27x}{x^2 - 10x + 25} dx$

$x^2 - 10x + 25 \overline{) x^3 - 10x^2 + 27x}$
 $-(x^3 - 10x^2 + 25x)$
 $0 + 2x$

$\int x dx + \int \frac{2x}{(x-5)^2} = \int x dx + \int \left(\frac{A}{x-5} + \frac{B}{(x-5)^2} \right) dx$

$A(x-5) + B = 2x$
 $x=5, B=10, A=2$

$Ax - 5A + B = 2x$

$= \frac{1}{2}x^2 + \int \frac{2}{x-5} + \frac{10}{(x-5)^2} dx$
 $= \frac{1}{2}x^2 + 2\ln|x-5| - \frac{10}{x-5} + C$

6. $\int \frac{2}{x(x^2+1)^2} dx$

$\int \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2} = 2\ln|x| - 2\ln|x^2+1| + \frac{1}{x^2+1} + C$
 $\ln\left(\frac{x^2}{x^2+1}\right) + \frac{1}{x^2+1} + C$

$2 = A(x^2+1)^2 + (Bx+C)(x^2+1) + (Dx+E)(x)$

@ $x=0, A=2$

$2 = A(x^4 + 2x^2 + 1) + (Bx+C)(x^3+x) + Dx^2 + Ex$

$2 = Ax^4 + 2Ax^2 + A + Bx^4 + Bx^2 + Cx^3 + Cx + Dx^2 + Ex$

$Ax^4 + Bx^4 = 0 \quad B = -2$

$2Ax^2 + Bx^2 + Dx^2 = 0 \quad D = -2$

$Cx^3 = 0 \quad C = 0$

$Cx + Ex = 0 \quad E = 0$

7. $\int \frac{x^2 - 2x + 5}{(x-3)(x^2+1)} dx$

$\int \frac{A}{x-3} + \frac{Bx+C}{x^2+1} = \int \frac{\frac{4}{5}}{x-3} + \frac{-\frac{1}{5}x - \frac{7}{5}}{x^2+1} = \frac{4}{5}\ln|x-3| + \frac{1}{10}\ln(x^2+1) - \frac{7}{5}\arctan x$

$A(x^2+1) + (Bx+C)(x-3) = x^2 - 2x + 5$

$x=3, 10A=8, A=\frac{4}{5}$

$x^2 - 2x + 5 = Ax^2 + A + Bx^2 - 3Bx + Cx - 3C$

$Ax^2 + Bx^2 = 0 \quad B = -\frac{1}{5}$

$3B + C - A = 0 \quad C = -\frac{7}{5}$
 $-\frac{3}{5} + \frac{4}{5} = \frac{1}{5}$

8. $y = \frac{10}{x^2 - 2x - 24}$, x-axis, $x = -2, x = 2$

$$\int_{-2}^2 \frac{10}{(x-6)(x+4)} dx = \int_{-2}^2 \frac{A \overset{1}{\cancel{x-6}}}{x-6} + \frac{B \overset{-1}{\cancel{x+4}}}{x+4}$$

$$A(x+4) + B(x-6) = 10$$

$$Ax + Bx = 0x$$

$$4A - 6B = 10$$

$$-10B = 10 \quad B = -1$$

$$Ax + 4A + Bx - 6B = 10$$

$$A + B = 0$$

$$4(-B) - 6B = 10$$

$$A = 1$$

$$\ln \left| \frac{x-6}{x+4} \right| \bigg|_{-2}^2 = \ln \left(-\frac{4}{6} \right) - \ln(4) = \ln \left| \frac{\frac{2}{3}}{4} \right| = \ln \left| \frac{1}{6} \right|$$

$$= \ln(6)^{-1} = -\ln 6 \leftarrow \text{under x-axis}$$

area is $\ln 6$